

Data Structures – Important Terms (Table)

Term	Meaning (Simple)	Example / Usage
Array	Collection of elements stored in contiguous memory.	<pre>int arr[5] = {1, 2, 3, 4, 5}</pre>
Linked List	A sequence of nodes where each node points to the next.	Singly list, doubly list
Stack	LIFO structure (last in, first out).	Undo functionality, parenthesis checking
Queue	FIFO structure (first in, first out).	Job scheduling, printers
Deque	Queue where insertion/deletion is allowed at both ends.	Browser history
Hash Table	Store key-value pairs using hashing for fast lookup.	Dictionaries, maps
Binary Tree	Each node has at most two children.	Expression trees
Binary Search Tree (BST)	Binary tree where $left < root < right$.	Searching sorted data quickly
Heap	Complete binary tree used for priority-based operations.	Priority queue, heap sort
Graph	Collection of nodes (vertices) and edges.	Networks, routes, maps
Adjacency Matrix	2D matrix representation of a graph.	Graph algorithms
Adjacency List	List-based graph representation.	Efficient for sparse graphs
Trie	Tree-like structure for string storage.	Auto-suggest, dictionary words
Dynamic Programming	Breaks problems into smaller overlapping subproblems.	Fibonacci, knapsack
Greedy Algorithm	Solves problems by picking the best current choice.	Kruskal, Prim
Recursion	Function calling itself to break problems into smaller parts.	Factorial, tree traversal
Big-O Notation	Measures time/space complexity of algorithms.	$O(n)$, $O(\log n)$
Sorting Algorithms	Techniques to arrange data in order.	Quick sort, merge sort

Term	Meaning (Simple)	Example / Usage
Searching Algorithms	Techniques to find an element.	Linear search, binary search
Collision Handling	Methods to resolve hash collisions.	Chaining, open addressing
Tree Traversal	Visiting nodes systematically.	Inorder, preorder, postorder